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Induction Homework

Part A

Mathematical Induction Proof Template

$\forall n \in \mathbb{N}\ (P(n))$

THEOREM: "For every $n \in \mathbb{N}$, $\underbrace{\boxed{1+2+\ldots+n = \frac{n(n+1)}{2}}}_{P(n)}$

PROOF: By mathematical induction.

State and prove P(1)

Basis: $P(1)$ asserts that $1 = \frac{1(1+1)}{2}$

which is true because $1 = \frac{1(2)}{2}$

$1 \overset{\checkmark}{=} 1$

Note: If appropriate, use P(0), P(2) or other value instead.

State $P(k)$ (inductive hypothesis)

Inductive step: Assume for an arbitrary $k \in \mathbb{N}$, $P(k)$ is true, i.e., namely:

$P(k) = 1+2+\ldots+k = \frac{k(k+1)}{2} = IH$

State $P(k+1)$

We will now show that $P(k+1)$ is also true, i.e.:

$1+2+\ldots+k+(k+1) = \frac{(k+1)((k+1)+1)}{2} = \frac{(k+1)(k+2)}{2}$

Prove $P(k) \Rightarrow P(k+1)$

Proof of inductive step:

$\underbrace{1+2+\ldots+k}+(k+1) = \frac{(k+1)(k+2)}{2}$

$\frac{k(k+1)}{2} + (k+1)$ $\boxed{IH}$

$= \frac{k(k+1)}{2} + \frac{2(k+1)}{2}$ $\boxed{\text{multiply by } \frac{2}{2}}$

$= \frac{k(k+1)+2(k+1)}{2} = \frac{k^2+k+2k+2}{2} = \frac{(k+1)(k+2)}{2}$

All done: wrap up proof

We thus have that $P(1)$ and $\forall k \in \mathbb{N}, P(k) \rightarrow P(k+1)$, so by the principle of mathematical induction, it follows that $P(n)$ is true for all natural numbers n.

Q.E.D.

Steps of a mathematical induction proof:
1) state the theorem, which is the proposition P(n)
2) show that P(base case) is true. Base case is usually P(1), but sometimes P(0) or P2) or other value is appropriate.
3) state the inductive hypothesis (substitute k for n)
4) state what must be proved (substitute k+1 for n)
5) state that you are beginning your proof of the inductive step, and proceed to manipulate the inductive hypothesis (which we assume is true) to find a link between the inductive hypothesis and the statement to be proven. Always state explicitly where you are invoking the inductive hypothesis.
6) finish your proof by invoking the principle of mathematical induction that allows you to infer that $P(n)$ is true for all natural numbers.

$n = \frac{n(n+1)}{2}$

Stuck on the proof of the inductive step? Do some examples for inspiration!

State the following and try to figure out why they are true. Then see if a pattern emerges that you can generalize.
Try some more base cases:

$P(2)$ $1+2 = \frac{2(2+1)}{2} \rightarrow 3 = \frac{2(3)}{2} \rightarrow 3 \overset{\checkmark}{=} 3$

$P(3)$ $1+2+3 = \frac{3(3+1)}{2} \rightarrow 6 = \frac{3(4)}{2} \rightarrow 6 \overset{\checkmark}{=} 6$

$P(4)$ $1+2+3+4 = \frac{4(4+1)}{2} \rightarrow 10 = \frac{4(5)}{2} \rightarrow 4 = 4$

Mathematical Induction Proof Template

$\forall n \in \mathbb{N}\ (P(n))$

THEOREM: "~~For every $n \in \mathbb{N}$,~~ odd $n \in \mathbb{N}$ $\boxed{\text{if } n \text{ is odd } 1+3+5+\ldots+(2n-1)=n^2}$

$P(n)$

PROOF: By mathematical induction.

State and prove $P(1)$

Basis: $P(1)$ asserts that $2(1)-1=1^2$

which is true because $2-1=1^2$

$1=1$ ✓

Note: If appropriate, use P(0), P(2) or other value instead.

State $P(k)$ (inductive hypothesis)

Inductive step: Assume for an arbitrary $k \in$ ~~$\mathbb{N}$~~ odd $\mathbb{N}$, $P(k)$ is true, i.e., namely:

$P(k): 1+3+5\ldots+(2k-1)=k^2$

We will now show that $P(k+1)$ is also true, i.e.:

State $P(k+1)$

$1+3+5\ldots+(2k-1)+(2(k+1)-1)=(k+1)^2$

$(2k-1)+(2k+1)=(k+1)^2$

Proof of inductive step:

Prove $P(k) \Rightarrow P(k+1)$

$1+3+5\ldots+(2k-1)+(2(k+1)-1)$

$=[1+3+5\ldots+(2k-1)]+2k+1$

$=k^2+(2k+1)$ $\boxed{\text{IH}}$

$=k^2+2k+1$ $\boxed{\text{algebra}}$

$=(k+1)^2$ $\boxed{\text{algebra}}$

All done: wrap up proof

We thus have that $P(1)$ and $\forall k \in \mathbb{N}, P(k) \rightarrow P(k+1)$, so by the principle of mathematical induction, it follows that $P(n)$ is true for all natural numbers n.

Q.E.D.

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6) finish your proof by invoking the principle of mathematical induction that allows you to infer that $P(n)$ is true for all natural numbers.

$2n-1=n^2$

Stuck on the proof of the inductive step? Do some examples for inspiration!

State the following and try to figure out why they are true. Then see if a pattern emerges that you can generalize.
Try some more base cases:

$P(2)$ $1+(2(2)-1)=(2)^2 \rightarrow 4=4$

$P(3)$ $1+3+(2(3)-1)=(3)^2 \rightarrow 9=9$

$P(4)$ $1+3+5+(2(4)-1)=(4)^2 \rightarrow 16=16$

Mathematical Induction Proof Template

$\forall n \in \mathbb{N}\ (P(n))$

THEOREM: "For every $n \in \mathbb{N}$, if $n \geq 0$ $\underbrace{1 + 2 + 2^2 + \ldots + 2^n = 2^{n+1} - 1}_{P(n)}$

PROOF: By mathematical induction.

State and prove P(1)

Basis: ~~P(1)~~ P(0) asserts that $2^0 = 2^{0+1} - 1$

$1 = 2 - 1$

which is true because $1 = 1$

Note: If appropriate, use P(0), P(2) or other value instead.

*State $P(k)$ (**inductive hypothesis**)*

Inductive step: Assume for an arbitrary $k \in \mathbb{N}$, $P(k)$ is true, i.e., namely:

$1 + 2 + 2^2 + \ldots + 2^k = 2^{k+1} - 1$

State $P(k+1)$

We will now show that $P(k+1)$ is also true, i.e.:

$1 + 2 + 2^2 + \ldots + 2^k + 2^{k+1} = 2^{(k+1)+1} - 1$

$= 2^{k+2} - 1$

Prove $P(k) \Rightarrow P(k+1)$

Proof of inductive step:

$(1 + 2 + 2^2 + \ldots + 2^k) + 2^{k+1}$

$= (2^{k+1} - 1) + 2^{k+1}$ $\boxed{\text{IH}}$

$= 2^{k+1} + 2^{k+1} - 1$ $\boxed{\text{associative}}$

$= 2^{k+2} - 1$ $\boxed{\text{add same bases}}$

All done: wrap up proof

We thus have that $P(1)$ and $\forall k \in \mathbb{N}, P(k) \rightarrow P(k+1)$, so by the principle of mathematical induction, it follows that $P(n)$ is true for all natural numbers n.

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$2^n = 2^{n+1} - 1$

Stuck on the proof of the inductive step? Do some examples for inspiration!

State the following and try to figure out why they are true. Then see if a pattern emerges that you can generalize.
Try some more base cases:

$P(2)$ $1 + 2 + 2^2 = 2^{2+1} - 1 \rightarrow 1 + 2 + 4 = 2^3 - 1 \rightarrow 7 = 7$

$P(3)$ $1 + 2 + 2^2 + 2^3 = 2^{3+1} - 1 \rightarrow 15 = 15$

$P(4)$ $1 + 2 + 2^2 + 2^3 + 2^4 = 2^{4+1} - 1 \rightarrow 31 = 31$

Mathematical Induction Proof Template

$\forall n \in \mathbb{N}\ (P(n))$

THEOREM: "For every $n \in \mathbb{N}$, $\underbrace{\boxed{\text{If } n \geq 0 \quad n < 2^n}}_{P(n)}$

PROOF: By mathematical induction.

State and prove P(1)

Basis: $P(1)$ asserts that $1 < 2^1$

$1 < 2$

which is true because

Note: If appropriate, use P(0), P(2) or other value instead.

Inductive step: Assume for an arbitrary $k \in \mathbb{N}$, $P(k)$ is true, i.e., namely:

State $P(k)$ (inductive hypothesis)

$k < 2^k$

We will now show that $P(k+1)$ is also true, i.e.:

State $P(k+1)$

$k+1 < 2^{k+1}$

Proof of inductive step:

Prove $P(k) \Rightarrow P(k+1)$

$k+1$

$< 2^k + 1$ $\boxed{1 \neq}$

$\leq 2^k + 2^k$ $\boxed{k \text{ may be } 0 \quad 2^0 = 1}$

$= 2 \cdot 2^k$ $\boxed{\text{simplify } (2^k)^2}$

$= 2^{1+2}$ $\boxed{\text{exponents to base } b}$

All done: wrap up proof

We thus have that $P(1)$ and $\forall k \in \mathbb{N}, P(k) \rightarrow P(k+1)$, so by the principle of mathematical induction, it follows that $P(n)$ is true for all natural numbers n.

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6) finish your proof by invoking the principle of mathematical induction that allows you to infer that $P(n)$ is true for all natural numbers.

$n < 2^n$ $n < 2^n$

Stuck on the proof of the inductive step? Do some examples for inspiration!

State the following and try to figure out why they are true. Then see if a pattern emerges that you can generalize.
Try some more base cases:

$P(2)$ $2 < 2^2 \rightarrow 2 < 4$

$P(3)$ $3 < 2^3 \rightarrow 3 < 8$

$P(4)$ $4 < 2^4 \rightarrow 4 < 16$

Mathematical Induction Proof Template

$\forall n \in \mathbb{N}\ (P(n))$

THEOREM: "For every $\cancel{n \in \mathbb{N}}$ $n \geq 4$, $\boxed{\text{if } n \geq 4 \quad 2^n < n!}$"

$P(n)$

PROOF: By mathematical induction.

State and prove $P(1)$

Basis: $\cancel{P(1)}$ $P(4)$ asserts that $2^4 < 4!$

$2^2 \cdot 2^2 < 4 \cdot 3 \cdot 2 \cdot 1$

which is true because $16 < 24$

Note: If appropriate, use P(0), P(2) or other value instead.

State $P(k)$ (inductive hypothesis)

Inductive step: Assume for an arbitrary $\cancel{k \in \mathbb{N}}$ $k \geq 4$, $P(k)$ is true, i.e., namely:

if $k \geq 4$

$2^k < k!$

State $P(k+1)$

We will now show that $P(k+1)$ is also true, i.e.:

$2^{(k+1)} < (k+1)!$

Prove $P(k) \Rightarrow P(k+1)$

Proof of inductive step:

2^{k+1}

$= 2^k \times 2^1$ $\boxed{\text{exp. to base b}}$

$< k! \times 2$ $\boxed{\text{IH}}$

$< k! \times (k+1)$ $\boxed{k \geq 4 \therefore 2 < k+1}$

$< (k+1)!$ $\boxed{\text{factorial } (k+1) \times k \times \ldots 1}$

All done: wrap up proof

We thus have that $P(1)$ and $\forall k \in \mathbb{N}, P(k) \rightarrow P(k+1)$, so by the principle of mathematical induction, it follows that $P(n)$ is true for all natural numbers n.

Q.E.D.

Steps of a mathematical induction proof:
1) state the theorem, which is the proposition P(n)
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6) finish your proof by invoking the principle of mathematical induction that allows you to infer that $P(n)$ is true for all natural numbers.

$2^n < n!$

Stuck on the proof of the inductive step? Do some examples for inspiration!

State the following and try to figure out why they are true. Then see if a pattern emerges that you can generalize.
Try some more base cases:

$\cancel{P(2)}$

$\cancel{P(3)}$

$P(4)$ $2^4 < 4!$

$2 \cdot 2 \cdot 2 \cdot 2 < 4 \cdot 3 \cdot 2 \cdot 1$

$16 < 24$

Part B

SECTION 2

1) $n \in \mathbb{N}$, $a^n \times b^n = (ab)^n$

STEP	P(x)	Proof
Base Case	P(1)	$a^1 \times b^1 = (ab)^1$
		$= a \times b = ab$ [exponents]
IH	P(k)	$a^k \times b^k = (ab)^k$ [multiply exponents]
induction	P(k+1)	$a^{k+1} \times b^{k+1} = (ab)^{k+1}$ [IH]
		$(a^k \times a^1)(b^k \times b^1) = (ab)^k \times (ab)^1$ [exp.]
		$= a^k \cdot b^k \cdot a^1 \cdot b^1$ [associative]
		$= (ab)^k \cdot (ab)^1$

2) $n \in \mathbb{N}$, $10^n \bmod 3 = 1$

STEP	P(x)	PROOF
Base	P(1)	$10^1 \bmod 3 = 1$
		$10 \bmod 3 = 1$
IH	P(k)	$10^k \bmod 3 = 1$
PROOF	P(k+1)	$10^{k+1} \bmod 3 = 1$
		$= (10^k \cdot 10^1) \bmod 3 = 1$ [mult. exp.]
		$= (10^k \cdot 10) \bmod 3 = 1$
		$= (10^k \bmod 3)(10 \bmod 3) \bmod 3 = 1$ [mod mult]
		$= (1)(1) \bmod 3 = 1$
		$= 1 \bmod 3 = 1$

3) $n \in \mathbb{N}$, $n \geq 1$

$$1^3 + 2^3 + \ldots + n^3 = \frac{n^2(n+1)^2}{4}$$

STEP	P(x)	PROOF
BASE	P(1)	$1^3 = \frac{1^2(1+1)^2}{4}$
		$1 = \frac{1^2(2)^2}{4}$
		$1 = 1$ ✓
IH	P(k)	$1^3 + 2^3 + \ldots + k^3 = \frac{k^2(k+1)^2}{4}$
PROOF	P(k+1)	$\underbrace{1^3 + 2^3 + \ldots + k^3}_{1} + (k+1)^3 = \frac{(k+1)^2((k+1)+1)^2}{4}$

$\boxed{\text{IH:}}$

$$\frac{k^2(k+1)^2}{4} + (k+1)^3 = \frac{(k+1)^2(k+2)^2}{4}$$

$\boxed{\text{multiply both sides by 4:}}$

$$\frac{\cancel{4}}{1} \cdot \frac{k^2(k+1)^2 + 4(k+1)^3}{\cancel{4}} = \frac{(k+1)^2(k+2)^2}{\cancel{4}} \cdot \frac{\cancel{4}}{1}$$

$$k^2(k+1)^2 + 4(k+1)^3 = (k+1)^2(k+2)^2$$

$\boxed{\text{factor } (k+1)^2}$

$$(k+1)^2\left[k^2 + 4(k+1)\right] = (k+1)^2(k+2)^2$$

$\boxed{\text{distribute 4}}$

$$(k+1)^2\left[k^2 + 4k + 4\right] = (k+1)^2(k+2)^2$$

$\boxed{\text{factor}}$

$$(k+1)^2(k+2)^2 = (k+1)^2(k+2)^2$$

Part C

The block video of the sum of values n can be a great visual aid to the induction proof. As in the video, the first part of the proof is to assess the theorem at a n= 1 (see below).

$1+2+\ldots+n = n(n+1)/2$

For the base case of n = 1 you would get the following:

$1 = 1(1+2)/2$

$1 = 1$

Similarly, for the block video, an n of 1 would include only 1 block. Therefore, the sum of the blocks in the "half the square" is 1.

If we assume that for any given value of k the theorem is true, using the block video, you can see that for any number of blocks in a column, the theorem would continue to hold true for the column directly to the right, that is, a column with size k + 1 blocks. The "staircase" could be added to indefinitely and the calculation of "half the square" would continue to equal the number of blocks used.